

Analysis III

Home Work 1

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Exercise 1 (Analysis II Recap, 1+1+1 P.)

Let $I = [a; b]$ be a compact interval in $\mathbb{R}$.

- (a) Explain the following statement: “The inclusion

$$T(I, \mathbb{R}) \subseteq S(I, \mathbb{R})$$

is dense with respect to the supremum-norm.” In particular, define the spaces $T(I, \mathbb{R})$ and $S(I, \mathbb{R})$ and the supremum-norm. (See Chapters 6 and 7 in the script.)

- (b) Explain briefly how the above can be used to define the (Riemann) integral

$$\int_a^b f(x)dx$$

for any $f \in S(I, \mathbb{R})$.

- (c) Construct a sequence of functions in $T(I, \mathbb{R})$ converging uniformly to $f(x) = x$ on $I = [0; 1]$. (You do not have to prove the uniform convergence.)

Exercise 2 (Null sets, 3+4 P.)

In this exercise we show that two sets are Lebesgue null.

- (a) Let $N = \mathbb{R}^{n-1} \times \{0\} \subset \mathbb{R}^n$. Using a similar strategy as in Example 11.1.4 (3) in the script, show that N is a set of measure zero.
- (b) Denote by

$$S^1 = \{x \in \mathbb{R}^2 : |x| = 1\}$$

the unit circle in $\mathbb{R}^2$. For $m \in \mathbb{N}$, take $x_1, \dots, x_m$ to be m equidistantly placed points on S^1 . Using squares of equal sizes centered at these points and eventually sending $m \rightarrow +\infty$, show that S^1 is a set of measure zero.

Hint: How large must these squares be so that they cover the circle? You may use without proof that the distance between two points on S^1 is at most as large as the angle between them.

ANSWER:

- (a) $N = \mathbb{R}^{(n-1)} \times \{0\}$ and now wish to show that this set in $\mathbb{R}^n$ has measure zero.

Given $\epsilon > 0$ consider the sequence of **open** and **bounded** intervals:

$$U_m := (-m, m) \times (-m, m) \times \dots \times (-m, m) \times \left(-\frac{\epsilon}{2^m(2m)^{(n-1)}}, \frac{\epsilon}{2^m(2m)^{(n-1)}}\right)$$

where there are $(n-1)$ terms of the form $(-m, m)$. Then it is clear that we have an open cover for N :

$$N \subset \bigcup_{m=1}^{\infty} U_m \tag{2.1}$$

We can easily compute the Lebesgue measure for each U_m as:

$$\lambda_n(U_m) = (2m)^{(n-1)} \frac{\epsilon}{2^m(2m)^{(n-1)}} = \frac{\epsilon}{2^m} \tag{2.1}$$

From which we conclude that:

$$\begin{aligned} \lambda_n(N) &\leq \sum_{m=1}^{\infty} \lambda_n(U_m) \\ &\leq \sum_{m=1}^{\infty} \frac{\epsilon}{2^m} \\ &\leq \epsilon \end{aligned}$$

As ϵ is arbitrary we can conclude that N is a set of measure zero (in $\mathbb{R}^n$).

(b) We will denote the set of points that comprise the boundary of the unit circle S^1 by ∂S .

Consider the diagram below showing the m points $p_1, p_2, \dots, p_m$ that are equally spaced on the circumference of the circle. The angle, extended from the centre of the circle, to each

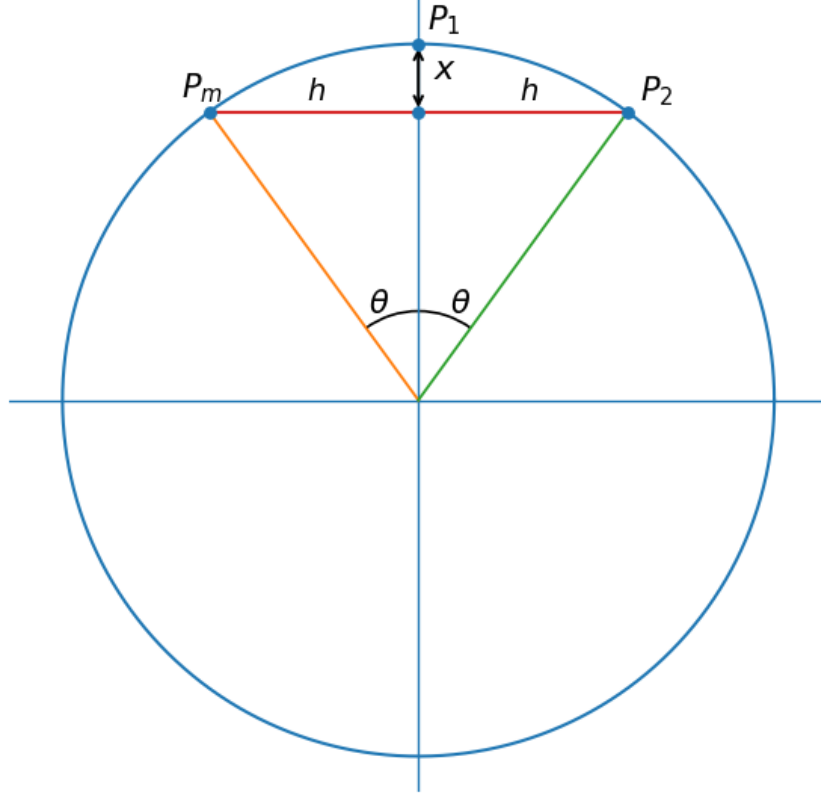


Figure 1: Circle construction

consecutive point on the circumference is $\theta = \frac{2\pi}{m}$. Now, for $m > 4$ we have that $h > x$ and also that $h < \frac{2\pi}{m}$ - which is the arc-length between any two neighbouring points on the circle's circumference.

Consequently, the point p_i can be contained in an **open** square of side length $2h = \frac{4\pi}{m}$, and we can refer to this open square as S_i .

It should be clear that the Lebesgue measure of each such set is: $\lambda_2(S_i) = (2h)^2 = \frac{16\pi^2}{m^2}$.

The m squares $(S_i)_{i=1}^m$ - which are **open** and **bounded** - overlap and completely cover the boundary, ∂S . Hence:

$$\partial S \subset \bigcup_{i=1}^{\infty} S_i \quad (2.3)$$

Hence, we can obtain an upper measure for the set ∂S from:

$$\begin{aligned}
\lambda_2(\partial S) &\leq \sum_{i=1}^m \lambda_2(S_i) \\
&\leq \sum_{m=1}^{\infty} \frac{16\pi^2}{m^2} \\
&\leq \frac{16\pi^2}{m}
\end{aligned} \tag{2.4}$$

Given any $\epsilon > 0$ we can find an integer $M \in \mathbb{N}$ such that $M > \frac{16\pi^2}{\epsilon}$ for which (2.4) is smaller than ϵ and we can conclude that, in $\mathbb{R}^2$, the set ∂S has measure zero.

Exercise 3 (Cantor set)

Recall the definition of the Cantor set $C = \bigcap_{n \in \mathbb{N}} C_n$ from Example 11.1.7 in the script: We define $C_0 = [0; 1]$ and proceed iteratively defining C_{n+1} from C_n by removing the middle third from every interval. Prove that

$$C = \left\{ \sum_{n=1}^{+\infty} a_n 3^{-n} : a_n \in \{0, 2\} \text{ for every } n \right\}.$$

Hint: You may use without proof that

$$[0; 1] = \left\{ \sum_{n=1}^{+\infty} a_n 3^{-n} : a_n \in \{0, 1, 2\} \text{ for every } n \right\}.$$

Exercise 4 (Equivalent definition of null sets)

Let $A \subset \mathbb{R}^n$. Show that A is a Lebesgue null set if and only if there exists a sequence of (n -dimensional) open intervals $(I_k)_{k \in \mathbb{N}}$ (in the sense of Definition 11.1.1) so that

$$\sum_{k \in \mathbb{N}} \lambda_n(I_k) < +\infty$$

and with the property that every $x \in A$ is contained in infinitely many I_k 's.

Exercise 5 (Jordan null sets)

Let $A \subset \mathbb{R}^n$. We say that A is a *Jordan null set* if for any $\varepsilon > 0$ there exist finitely many (n -dimensional) open intervals $I_1, \dots, I_m$ with the property

$$\sum_{k=1}^m \lambda_n(I_k) < \varepsilon \quad \text{and} \quad A \subset \bigcup_{k=1}^m I_k.$$

Observe that by definition every Jordan null set is also Lebesgue null.

- (a) Show that any Jordan null set is necessarily bounded.
- (b) Give an example of an uncountable Jordan null set in $\mathbb{R}^2$.
- (c) Give an example of a Lebesgue null set in $\mathbb{R}$ that is *not* a Jordan null set.

Exercise 6 (Dini's Theorem)

Let $M = [a; b]$ be a compact interval (or, more generally a compact set). Given is a sequence of continuous functions

$$f_n : M \rightarrow \mathbb{R}$$

that is decreasing, i.e. for every $n \in \mathbb{N}$ and $x \in M$ we have $f_n(x) \geq f_{n+1}(x)$. Assume moreover that there is another continuous function $f : M \rightarrow \mathbb{R}$ so that $f_n \rightarrow f$ pointwise.

(a) Prove that $f_n \rightarrow f$ uniformly. In order to do this, consider the sets

$$E_n = \{x \in M : f_n(x) - f(x) < \varepsilon\}, \quad n \in \mathbb{N}$$

and use compactness.

(b) Show that the continuity of f is necessary.